Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/06/2025
PI Name:	Zerdoum, Aidan		
Project Title:	Fundamentals of Ultrasound - Artemis Training Solution		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2025 per TRISH (Ed., 9/27/24).		
Key Personnel Changes/Previous PI:	Per TRISH, the PI changed from Victoria Perizes to Aidan Zerdoum. The first NASA Task Book reporting is from Dr. Zerdoum (Ed., 10/1/24).		
COI Name (Institution):	Sargsyan, Ashot M.D. (KBR)		
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Task Description:	SUMMARY Level Ex will build upon the successful completion of the Virtual Human Simulation (VHS) project to create Fundamentals of Ultrasound - Artemis Training Solution in support of the Translational Research Institute for Space Health (TRISH) mission to identify and promote disruptive, breakthrough approaches that reduce risks to human health and performance. Teams from Level Ex and KBR will work with TRISH and NASA stakeholders to create, and evaluate in a human-in-the loop testing scenario, an interactive ultrasound training solution. This solution aims to provide experiential medical practice and will study the efficacy of our interactive training solution as compared to traditional didactic ultrasound procedural training. Collectively, these aims support addressing Risks and Gaps identified in the HRP Human Research Roadmap for crew training and personalized semi-autonomous and autonomous health care during future human exploration missions. Specific Aims: 1. Create an interactive game / task training solution, which introduces users to the fundamental aspects of ultrasound image acquisition and provides a means to practice the necessary skills required to acquire high-quality image sets. 2. Evaluate the efficacy of the newly designed interactive game, Fundamentals of Ultrasound - Artemis Training Solution, against a traditional didactic-based training modality using a randomized, controlled comparative study. Key Findings: The key findings will affirm the project's hypothesis that a game-based approach can serve as an effective teaching tool for ultrasound procedures. The incremental spatial cognition tests, designed to mimic real-world complexities, facilitate cognitive assimilation and spatial reasoning, enhancing educational rigor. Modules focusing on diagnostic accuracy further legitimize the game's educational value. User-centric features such as the progression header and real-time indicators like breathing cadence and probe pressure engage the user while also serving pedagogical purposes. High-fidelity 3D models and practical skill acquisition tools, like the self-scanning hand model, bridge the gap between theory and practice, making the training comprehensive and aligned with real-world clinical skills. Finally, improvements to existing ultrasound simulation technology result in a realistic representation of ultrasound imagery. These features collectively contribute to the game’s efficacy as a comprehensive, engaging, and practical platform for teaching ultrasound procedures. Research Plan: The next phase of the project will focus on empirically validating the effectiveness of the Fundamentals of Ultrasound - Artemis Training Solution through a rigorous, randomized, controlled comparative study against traditional didactic-based training. This efficacy study aims to generate quantifiable metrics that assess both cognitive assimilation and practical skill acquisition. Data from this study will not only quantify the game's instructional efficacy but also inform iterative refinements to optimize its educational value. Following the analysis of these findings, a scholarly publication will be prepared to disseminate the research outcomes, thereby contributing to the broader discourse on innovative pedagogical approaches in medical training.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Artemis Training Solution has potential applications beyond space medicine, including remote and resource-limited environments on Earth. By enhancing cognitive assimilation and spatial reasoning, the platform can help train first responders, military medics, and clinicians in developing regions where real-time medical support is not available.
Task Progress:	The Fundamentals of Ultrasound - Artemis Training Solution project aims to develop and evaluate an interactive training platform for autonomous ultrasound training in spaceflight settings. The primary goals of this project are to train a group of lay-users in basic ultrasound skills (Aim 1), then validate the efficacy of this training compared to traditional methods (Aim 2). Year 1 focused on initial game development, including the creation of 3D anatomical models and interactive training modules. Years 2+ were mainly focused on completion of the randomized controlled study evaluating the efficacy of medical training software in producing quality clinical imaging data. SUMMARY OF KEY FINDINGS AND MILESTONES • Developed a high-fidelity ultrasound training game with modules covering probe handling, image acquisition, anatomical orientation, and diagnostic image interpretation. • Conducted a randomized controlled trial comparing training methodologies. Preliminary results indicate that interactive medical training software can effectively train lay users to acquire diagnostic ultrasound images. • Integrated telemetry data analysis to assess user performance, including module completion rates, average time to task, and multiple choice question (MCQ) accuracy. • Collected procedural image scores for the internal jugular vein (IJV) task. PROGRESS TOWARD SPECIFIC AIMS • Aim 1: Fully developed interactive training modules including: probe positioning, neck anatomy, and Doppler imaging. Incorporated relevant anatomical models and context specific tasks to reinforce real world scanning techniques. • Aim 2: Progressing towards completion of the efficacy study. Ongoing data analysis includes procedural image scoring and thematic analysis of participant feedback. In Year Two, the primary focus has been the execution and completion of Aim 2, which involves conducting a randomized trial to assess participant performance on an imaging task. This includes finalizing data collection from the remaining enrolled subjects and performing comprehensive data analysis. The study will evaluate group level performance metrics relating to the diagnostic quality of the acquired images. Additionally, participant feedback will be incorporated into a human factors analysis. Outcomes will focus on the overall efficacy and interoperability between interactive training software and the current NASA training paradigm.
Bibliography Type:	Description: (Last Updated:)